EEE 495 Electrical and Electronics Engineering Design II

Test Plan Document Draft

Project Title / Product Name: FarmGate Sentinel

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Team No: 2

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1 Introduction

1.1 Purpose of the Test Plan

This Test Plan document defines the methods, procedures, and acceptance criteria that will be used to verify that the *FarmGate Sentinel* system satisfies the functional and non-functional requirements specified in the Project Specifications Document. The purpose of this document is to ensure that all critical system components operate correctly, reliably, and within the defined performance limits.

The tests described in this document are designed to validate measurable system behaviors under realistic operating conditions. Successful execution of these tests will demonstrate that the system meets its intended design objectives, including reliable wildlife detection, timely alert generation, secure communication, and effective deterrence mechanisms.

1.2 Product Overview

FarmGate Sentinel is a vision-first Internet of Things (IoT) system designed to detect and deter wildlife intrusions near agricultural gates, corrals, and fence lines. The system consists of distributed edge camera nodes equipped with visual and environmental sensors, a central gateway unit, cloud-based services, and a mobile application for user interaction.

Edge nodes continuously monitor their surroundings by capturing images and sensor data. Upon detecting suspicious activity, the nodes transmit the collected data to the gateway unit via a local wireless connection. The gateway processes incoming data, performs detection and incident correlation, and communicates with cloud services through an LTE connection. When a confirmed event is detected, users are notified through a mobile application and may activate deterrence mechanisms such as an audible buzzer.

1.3 Scope of Testing

The scope of this Test Plan includes verification of both functional and non-functional requirements of the FarmGate Sentinel system. Functional tests focus on correct operation of individual subsystems and their integration, including image capture, data transmission, detection latency, alert delivery, and user interaction through the mobile application. Non-functional tests address performance constraints, power consumption, communication reliability, environmental tolerance, and interface robustness.

Only the specifications that will be demonstrated and verified during the final project evaluation are included in this Test Plan.

2 Functional Requirements[1]

Labels are taken from the technical specifications document report.We realized that some functional requirements of ours are actually non-functional requirements therefore they are moved to non-functional requirements section.

2.1 Edge Node Tests

- **Scope:** FR-EDGE-01 to FR-EDGE-06.FR-EDGE-07 is moved to non-functional requirements

- **Test setup / equipment:**
  - Fully assembled Edge Node (ESP32-CAM + sensors + status LED + buzzer + enclosure).
  - Wi-Fi access point and backend/logging destination running.
  - Laptop for policy changes and log collection; stopwatch/timer; phone camera; ruler measure.
- **Test plan:**
- **Responsible Team Member**: Muhammed Gölcük is responsible for all tests in edge node functional requirements part.

1. **TN-EDGE-PRE01: Detection Distance**
   - **Requirement(s):**
     * *For a full size animal the distance is five meters but for lab demonstration a tablet size is a required for 80cm at least.*
   - **Method / procedure:**
     * a photograph of the chosen animal will increment step by step shown through camera for each step we will increase the distance by 40 cm and test till 80 cm and then we will increment the distance by one meter till system doesn't recognize the animal.
   - **Acceptance criteria:**
     * We will accept this requirement successful if it works at least 80 cm away from camera.
2. **TN-EDGE-01: Capture schedule and policy update**
   - **Requirement(s):**
     * *"The node shall capture images at a **baseline schedule** of **10 frames/minute** with tolerance* ± 1 *frame/min, and support a policy-configurable range of **2–30 frames/min** without reflashing."*[1] Due to hardware circumstances it is now in range of 2 to 15 frames/minute without refreshing.
   - **Method / procedure:**
     * Baseline capture policy is set to 10 frames/minute (fpm).
     * The node is operated for 3 minutes while image timestamps and total frame count are observed.
     * The time until the new rate is applied is recorded; timestamps are logged for an additional verification window.
   - **Acceptance criteria:**
     * For 3 minutes at 10 fpm, total frames are within $[27, 33]$.
3. **TN-EDGE-02: Resolution, JPEG quality, and file size**
   - **Requirement(s):**
     * *"Default resolution shall be **800×600** (SVGA) with JPEG quality 50±5%; .High-detail mode will be **1280 ×1024** .In standard mode, 1.44MB will be the average file size. [1] This requirement has changed because we didn't realize that the average file size for SVGA, or 1280 x 1024 (SXGA), doesn't exceed 40 KB."*

– **Method / procedure:**
  - Standard mode is now configured (1280×1024).
  - 10 frames are captured; file sizes are recorded; headers are inspected to confirm resolution.
  - 10 frames are captured; file sizes are recorded; headers are inspected to confirm resolution.

– **Acceptance criteria:**
  - This test meets criteria when we receive 10 1280X1024 size frames

4. **TN-EDGE-03: Relocation / integrity regulation alert**

– **Requirement(s):**
  - *"The edge module is able to regulate its integrity. In the event of relocation or replacement, the user will be notified."*[1]

– **Method / procedure:**
  - The edge module is placed at a stable reference location and normal operation is confirmed.
  - The module is displaced from the stable location (simulating relocation/replacement or a fall from the determined height).
  - User-facing notifications and system logs are inspected for an alert event.
  - Defined alerts are tagged Object Moved and Freefall. The state object is added as an extra to determine object relocation

– **Acceptance criteria:**
  - A notification/alert is generated when displacement is performed.

5. **TN-EDGE-04: Camera obstruction / near-object detection**

– **Requirement(s):**
  - *"The system is able to detect moving objects in front of it in the event of camera obstruction. It will detect objects closer than 1 meter."*[1]

– **Method / procedure:**
  - Objects are placed in front of the camera at distances from 0 cm to 1 m in 10 cm increments.
  - For each distance, the detection outcome is recorded.
  - Beyond 1 m, placement is continued in 50 cm increments to characterize behavior above the threshold.

– **Acceptance criteria:**
  - Objects closer than 1 m are detected as required during the stepped-distance test.This operation is done by PIR sensor.

6. **TN-EDGE-05: Status LED state indication**

– **Requirement(s):**
  - *"LED shall indicate states: **slow blink** (Wi-Fi joining), **solid** (online), **fast blink** (burst)."*

– **Method / procedure:**
  - Wi-Fi joining state is entered and a 60 s LED video is recorded.
  - Online/connected state is confirmed and a 60 s LED video is recorded.

* Burst state is triggered and a 60 s LED video is recorded.
  * Videos are analyzed to confirm blink patterns and approximate rates.
- **Acceptance criteria:**
  * Slow blink is observed during Wi-Fi joining.
  * Solid LED is observed when online.
  * Fast blink is observed during burst.

7. **TN-EDGE-06: Buzzer actuation latency and on-time**
   - **Requirement(s):**
     * *"On `actuate:buzzer` command, an active buzzer shall sound with **start latency** $\leq$ **1 s**. Default on-time: **20 s** $\pm$ **1 s**."*
   - **Method / procedure:**
     * The `actuate:buzzer` command is issued 10 times.
     * Start latency and on-time duration are measured for each activation.
   - **Acceptance criteria:**
     * Start latency is $\leq$ 1 s for each trial.
     * On-time duration is 20 s $\pm$ 1 s.
     * Which means after predator detection we receive the buzzer alert at t=21s at most.

2.2 Gateway Node Tests

Responsible Team Member: Yusuf Semih Gürkok is responsible for all tests in gateway node functional requirements part.

- **Scope:** FR-GW-01 to FR-GW-03.
- **Test setup / equipment:**
  - Gateway Node with detection software and logging enabled.
  - Edge Node capable of sending SXGA frames to the gateway .
  - WAN link that can be disabled/enabled.
- **Test plan:**
  1. **TN-GW-01: Detection processing latency**
     - **Requirement(s):**
       * *"From image receipt to incident proposal, processing shall be $\leq$ **2.0 s** (p85) at SVGA."*[1]This requirement has been upgraded since we decided to process at an upper resolution SXGA.
     - **Method / procedure:**
       * A set of 25 labeled SXGA frames is fed to the gateway (live from node or replay).
       * For each frame, the time from *image receipt* to *incident proposal* is recorded from logs/timestamps.
       * The p85 processing latency is computed.

- **Acceptance criteria:**
  * The p85 processing latency is ≤ 2.0 s.

2. **TN-GW-02: WAN-down local buffering and drain**
   - **Requirement(s):**
     * *"With WAN down, gateway shall buffer ≥ **1000 images** and drain with ≥ **85%** success within ≤ **30 min** after link returns."*[1]. This requirement has been updated since we dont have 30 minutes to demonstrate this therefore. We will test this by 1 minute of buffering then draining the buffered images in 5 minutes.
   - **Method / procedure:**
     * WAN connectivity is disabled (.
     * 10 images are sent from a node to the gateway and are allowed to accumulate in the local buffer.
     * WAN connectivity is restored.
     * Drain duration is recorded and success ratio is computed as:

$$\text{Drain Success} = \frac{\text{Images uploaded/forwarded successfully}}{\text{Images buffered}}$$

   - **Acceptance criteria:**
     * At least 10 images are buffered while WAN is down.
     * Drain success is $\geq 85\%$.
     * Drain completes within ≤ 5 min after WAN restoration.

2.3 Communications Tests (Node ↔ Gateway ↔ Cloud)

Responsible Team Member: İsmail Özgenç is responsible for all tests in communications' functional requirements part.

- **Scope:** FR-COMM-01 to FR-COMM-04.
- **Test setup / equipment:**
  - One Edge Node and one Gateway Node with Wi-Fi 2.4 GHz enabled.
  - LTE 4.5G modem with SIM card installed (WAN path), plus a way to disable/restore WAN.
  - Logging enabled on node and gateway; timestamps embedded in payloads when required.
  - Tape measure for distance testing.
- **Test plan :**
  1. **TN-COMM-01: InterSystem communication methods**
     - **Requirement(s):**
       * *"Node-to-gateway communication method is Wi-Fi 2.4GHz, and the gateway communication method is a LTE 4.5G Modem with a SIM card that the user must purchase."*[1]

– **Method / procedure:**
  * The Edge Node is configured to connect to the Gateway over 2.4 GHz Wi-Fi.
  * The Gateway WAN uplink is provided through the LTE 4.5G modem with an active SIM.
  * Connectivity is confirmed by verifying that images/messages are received at the gateway and forwarded to the cloud when WAN is available.

– **Acceptance criteria:**
  * Node-to-gateway traffic is carried over Wi-Fi 2.4 GHz.
  * Gateway-to-cloud traffic is carried over the LTE 4.5G modem uplink.

2. **TN-COMM-02: Node Wi-Fi minimum range and quality downgrade**
   – **Requirement(s):**
     * *"Node shall maintain Wi-Fi to the gateway within at least 5m. In the event of a poor connection, the system will switch to a lower-quality image capture."*
   – **Method / procedure:**
     * The node and gateway are separated starting at 0 m.
     * Separation is increased by 50 cm per step up to 10 m, and connectivity is checked at each step.
     * At poor-connection steps (as indicated by RSSI/packet loss/throughput threshold used by the system), the capture quality mode is observed in logs (resolution/quality downgrade event).
   – **Acceptance criteria:**
     * A maintained Wi-Fi link is demonstrated for distances up to at least 5 m.
     * When a poor connection is detected, switching to a lower-quality capture mode is observed in logs and/or output images.

3. **TN-COMM-03: Capture-to-gateway receipt latency**
   – **Requirement(s):**
     * *"Capture-to-gateway receipt latency shall be ≤* ***500 ms*** *(p85) at SVGA; ≤* ***1.0 s*** *(p85) at VGA."*[1]
   – **Method / procedure:**
     * Capture timestamps are embedded at the node (or captured at the moment of send).
     * Receipt timestamps are recorded at the gateway upon full payload receipt.
     * Latency distributions are computed over 60 frames at SVGA and 60 frames at VGA.
     * The p85 latency is computed for both modes.
   – **Acceptance criteria:**
     * SVGA: p85 latency is ≤ 500 ms.
     * VGA: p85 latency is ≤ 1.0 s.
     * SXGA: p85 latency is ≤ 2.0 s.

4. **TN-COMM-04: Node→gateway delivery reliability**
   – **Requirement(s):**
     * *"Successful delivery node→gateway shall be ≥* ***90%*** *over 5 minutes at 10 fpm."*[1]

- **Method / procedure:**
  - The node is operated at 10 fpm for 5 minutes.
  - Images sent by the node are counted and compared to images received by the gateway.
  - Delivery success ratio is computed as:

$$\text{Delivery Success} = \frac{\text{Received at gateway}}{\text{Sent by node}}$$

- **Acceptance criteria:**
  - Delivery success ratio is ≥ 90% over the 5-minute test interval.

2.4 Cloud & Alerting Tests

Responsible Team Member: Zeynep İrem İpek is responsible for all tests in cloud and alerting functional requirements part.

- **Scope:** FR-CLOUD-01 to FR-CLOUD-04.
- **Test setup / equipment:**
  - Test cloud environment (staging) with logging and a test storage bucket configured.
  - Two node identity sources (two edge nodes or a replay harness that simulates two nodes).
  - Mobile client(s) registered for push notifications.
  - Test accounts/roles and tokens to validate RBAC enforcement.
- **Test plan:**
  1. **TN-CLOUD-01: Correlation into incidents**
     - **Requirement(s):**
       - *"Cloud shall merge detections into incidents when start times are < **10 s** apart and locations within ≤ **10 m**."*[1]. This requirement has been changed to "Cloud shall merge detections into incidents when start times are < **1 min** apart."
     - **Method / procedure:**
       - Two detection streams are replayed such that $\Delta t = 20$ s and $\Delta d = 5$ m.
       - Incident records are inspected to confirm a single merged incident is created.
       - Replay is repeated with $\Delta t = 80$ s (keeping other parameters comparable).
       - Incident records are inspected to confirm separate incidents are created.
     - **Acceptance criteria:**
       - With $\Delta t = 20$ s and $\Delta d = 5$ m, exactly one incident is produced.
       - With $\Delta t = 80$ s, separate incidents are produced.
  2. **TN-CLOUD-02: Incident storage and retention**
     - **Requirement(s):**

    * *"For each incident store ≥ **1 thumbnail** and ≥ **1 original** frame/clip with retention ≥ **7 days**."*[1]
  - **Method / procedure:**
    * A test incident is created (via replay or real detection).
    * Storage keys/objects are queried to confirm at least one thumbnail and one original are stored.
    * The same objects are re-fetched after 7 days in the test bucket/environment.
  - **Acceptance criteria:**
    * At least one thumbnail and one original are present per incident.
    * Objects remain retrievable after 7 days.

3. **TN-CLOUD-03: Push alert delivery latency**
  - **Requirement(s):**
    * *"On incident confirmation, push notification to registered clients within ≤ **30 s** (p85) of gateway detection."*[1]
  - **Method / procedure:**
    * Incident confirmations are generated for 5 trials.
    * Gateway detection timestamps and mobile receipt timestamps are recorded.
    * The p85 of $\Delta t = t_{\text{mobile receipt}} - t_{\text{gateway detection}}$ is computed.
  - **Acceptance criteria:**
    * Push notification p85 latency is ≤ 30 s.

4. **TN-CLOUD-04: RBAC enforcement and audit logging**
  - **Requirement(s):**
    * *"Enforce user roles. Unauthorized actions return HTTP 403 auth fail and are audit-logged."*[1]
  - **Method / procedure:**
    * Protected API calls are attempted using downgraded/unauthorized tokens.
    * API responses are recorded.
    * Audit logs are inspected for entries corresponding to the unauthorized attempts.
  - **Acceptance criteria:**
    * HTTP 403 responses are returned for unauthorized API actions.
    * Audit log entries are present for the unauthorized attempts.

2.5 User Interface (Web / Mobile) Tests

Responsible Team Member: Zeynep İrem İpek is responsible for all tests in edge node functional requirements part.

- **Scope:** FR-UI-01 to FR-UI-03.
- **Test setup / equipment:**
  - Web UI and mobile app builds connected to the test backend.
  - Test dataset containing 3 incidents.

– We will test two accounts: *Admin* and *Non-admin* .
– At least one active node for live view refresh tests .
– Timing method for p95 measurement.

- **Test plan:**

1. **TN-UI-01: Incident cards and account management**
   - **Requirement(s):**
     * *"UI shall display incident cards with **thumbnail**, **time**, **node ID**, **site**, **controls** buzzer, and the mobile app will allow creating, editing, and changing privileges of accounts"*[1]
   - **Method / procedure:**
     * The mobile app is opened and the incident list view is navigated to.
     * Incident cards are opened for test incidents and required fields are inspected.
     * Buzzer control is exercised from the incident card and the control response is observed.
     * Account management screens are opened.
     * A new account is created, an existing account is edited.
     * Admin/non-admin behavior is checked by repeating privileged actions using the non-admin account.
   - **Acceptance criteria:**
     * Incident cards display: thumbnail, time, node ID, site, and buzzer control.
     * Buzzer control is responsive and produces the expected system action (UI feedback and/or device behavior/log).
     * Account creation and editing are completed successfully in the app.
     * Privilege/role changes are applied as expected (privileged actions are restricted for non-admin).
2. **TN-UI-02: Live view refresh latency**
   - **Requirement(s):**
     * *"On request, UI shall fetch latest available image per node within ≤ **10 s** (p95)."*[1]
   - **Method / procedure:**
     * The live view screen is opened for a selected node.
     * The refresh action is triggered 5 times.
     * See if the last images are played.
   - **Acceptance criteria:**
     * The p95 refresh-to-render time is ≤ 10 s.
3. **TN-UI-03: Policy editing and propagation**
   - **Requirement(s):**
     * *"Admin can update per-node schedule (fpm), by burst capture, and actuator; changes take effect within ≤ **60 s**."*[1]
   - **Method / procedure:**
     * The admin account is used to log into the UI.

        * Starting the actuator by start is tested.
        * Node/gateway behavior and logs are observed to detect when the new policy takes effect.
        * The elapsed time from UI save/submit to observed behavior shift is recorded for each policy change by observation.
    - **Acceptance criteria:**
        * Schedule, burst threshold, and actuator duration updates are accepted and saved by the UI.
        * Each change is observed to take effect at the node/gateway within $\leq 60\,\text{s}$.

3 Non Functional Requirements Tests

Ambient test temperature is taken as 20 ± 5°C; supply is held at nominal; values are for a two–edge-node deployment.

3.1 Edge Node

Responsible Team Member: Muhammed Gölcük is responsible for all tests in edge node non-functional requirements part.

- **Environmental operating conditions**
    - **Requirement(s):**
        * *"Operating temperature **-10°C → 50°C**; humidity ≤ **95%** non-condensing; light rain splash tolerated by enclosure.[1]"*
    - **Method / procedure:**
        * Temperature exposure across -10°C to 50°C is performed, while capture/communication is monitored.
        * Unfortunately, can't test humidity.
        * The enclosure is sprayed lightly to simulate rain splash, and we look inside to see if any water is present.
    - **Acceptance criteria:**
        * Enclosure is tested. The Raspberry Pi already heats up above 50 degrees by itself.

3.2 Node Power Tests

Responsible Team Member: Muhammed Gölcük is responsible for all tests in node power tests non-functional requirements part.

- **Scope:** FR-PWR-01 to FR-PWR-04.
- **Test setup / equipment:**
    - Intended 5V adapter and/or bench power supply with adjustable current limit.
    - Fuse installed as in final design.

- **Test plan:**

  1. **TN-PWR-01: 5V input verification**
     - **Requirement(s):**
       - *"System operates at 5V DC source. Adapter will be take input 220V DC and convert it to 5V DC for ESP32[1]."*
     - **Method / procedure:**
       - Will be tested on fuse by giving fuse more than 1A power to see if fuse interrupts power.
     - **Acceptance criteria:**
       - Stable node operation is observed while 5V input is applied and confirmed by measurement.
  2. **TN-PWR-02: Overcurrent protection / fuse trip**
     - **Requirement(s):**
       - *"There will be fuses in place to protect the system electronics in the event of an electric spike or overflow, stopping currents larger than 1A [1]."*
     - **Method / procedure:**
       - The node is powered through a bench supply with monitoring enabled.
       - A controlled overflow condition is applied such that current demand exceeds 1A.
       - Fuse operation (current interruption) and post-test node condition are documented.
     - **Acceptance criteria:**
       - Currents larger than 1A are interrupted by the protection mechanism and electronics remain operational afterward.
  3. **TN-PWR-03: Reverse polarity survival**
     - **Requirement(s):**
       - *"Node shall survive **reverse polarity** on input without permanent failure[1]."*
     - **Method / procedure:**
       - Reverse polarity is applied to the input for 5 s.
       - Correct polarity is restored and normal power is applied.
       - Boot and basic functional operation are verified (Wi-Fi connect, capture, and/or actuation).
     - **Acceptance criteria:**
       - No permanent failure is observed; normal operation is recovered after polarity is corrected.
  4. **TN-PWR-04: Average current limits (baseline and idle)**
     - **Requirement(s):**
       - *"At baseline 10 fpm, average node current shall be ≤ **360 mA**. In idle (no capture, Wi-Fi connected): ≤ **240 mA**[1]."*
     - **Method / procedure:**
       - Baseline mode is configured (10 fpm) and current is logged for 2 minutes; mean current is computed.

- * Idle mode is configured (no capture, Wi-Fi connected) and current is logged for 2 minutes; mean current is computed.
- **Acceptance criteria:**
  - * Baseline mean current is ≤ 360 mA.
  - * Idle mean current is ≤ 240 mA.

3.3 Gateway Nonfunctional Requirements

Responsible Team Member: Yusuf Semih Gürkök is responsible for all tests in gateway non-functional requirements part.

- **TN-GW-03: Gateway environmental operation**
  - **Requirement(s):**
    - * *"Operating temperature **-10°C → 50°C**; humidity ≤ **95%** non-condensing; light rain splash tolerated by enclosure[1]."*
  - **Method / procedure:**
    - * Temperature exposure across -10°C to 50°C is performed if there are any equipments are available to perform that
  - **Acceptance criteria:**
    - * Continuous gateway operation is maintained within stated limits.

3.4 Size and Weight of Physical Sub-Components

Responsible Team Member: Yusuf Semih Gürkök is responsible for all tests in size and weight of physical sub-components non-functional requirements part.

- **Verification approach:** Inspection using a digital scale and ruler/calipers. Results are recorded and compared to the stated limits.
- **Test plan:**
  1. **TN-NFR-SW-01: Edge node mass check**
     - **Requirement(s):**
       - * *Each **edge node** weighs approximately **500 g** with enclosure with tolerance of ± 200g*
     - **Method / procedure:**
       - * Each assembled edge node is weighed including its enclosure.
       - * The measured mass is recorded for each unit.
     - **Acceptance criteria:**
       - * Measured mass per edge node is within 500–550 g (±200g deviation is justified).
  2. **TN-NFR-SW-02: Gateway unit mass check**
     - **Requirement(s):**
       - * *"The **gateway unit** weighs around **200-250 g**, suitable for mounting on a fixed positions."*

- **Method / procedure:**
  * The assembled gateway unit is weighed (enclosure + gateway computer + modem if included in the unit definition).
  * The measured mass is recorded.
- **Acceptance criteria:**
  * Measured gateway mass is within 450–550 g.

3. **TN-NFR-SW-03: Total deployed mass check**
   - **Requirement(s):**
     * *"The total deployed system (gateway + two edge nodes) remains under **2 kg**, ensuring portability and low-impact mounting in rural environments."*
   - **Method / procedure:**
     * Masses of the gateway unit and two edge nodes are summed (or the full set is weighed together).
     * The total mass is recorded.
   - **Acceptance criteria:**
     * Total deployed mass is < 2 kg.

3.5 Power Supply Specifications

Responsible Team Member: Muhammed Gölcük is responsible for all tests in power supply non-functional requirements part.

- **Verification approach:** Test by multimeter and current measurement. Analysis is used where power is computed from measured V/I.
- **Test plan:**

1. **TN-NFR-PWR-01: Gateway supply capability check**
   - **Requirement(s):**
     * *"The **gateway** is continuously powered from an official 5 V 3 A USB-C adapter to ensure stable operation under high load[1]."*
   - **Method / procedure:**
     * The gateway is powered using the specified 5 V 3 A adapter.
     * Operation is observed while CPU and USB load are applied (systems running).
     * Input voltage at the gateway is measured during the load period.
   - **Acceptance criteria:**
     * Stable operation is maintained under the applied load without reboot/brownout symptoms.
2. **TN-NFR-PWR-02: Voltage tolerance check**
   - **Requirement(s):**
     * *"Voltage tolerances are maintained within ± 5 %, protecting Edge Computer and sensor circuitry[1]."*
   - **Method / procedure:**

    * Edge supply rails (3.3 V) are measured at the edge node with multimeter under typical operation.
    * Measurements are repeated during peak activity (capture + Wi-Fi transmit + sensors active).
  – **Acceptance criteria:**
    * Measured rails remain within ±5% of nominal.

3. **TN-NFR-PWR-03: Edge-node average power estimate**
  – **Requirement(s):**
    * *"Average edge-node power consumption is below 1.3 W, allowing extended duty cycles with sleep modes[1]."*
  – **Method / procedure:**
    * Average current is measured for operating period.
    * Average power is computed as $P_{\text{avg}} = V \times I_{\text{avg}}$ using measured voltage and measured average current.
  – **Acceptance criteria:**
    * Computed $P_{\text{avg}} < 1.3$ W (or operating mode is documented if above).

3.6 Software Interface and Protocol Tests

Table 1: Non-Functional Tests for Software Interface / Communication Specifications

Test ID	Requirement(s)	Method / Procedure (incl. Acceptance)	Resp.
TN-NFR-SW-01	*Wi-Fi link shall transmit images/audio/sensor readings from edge to gateway.*	The edge node shall be connected to the gateway network and HTTP transfer shall be initiated. Mixed payloads (JPEG + JSON + optional WAV) shall be transmitted for 5 minutes. **Acceptance:** Transfers shall complete without protocol errors and received payload integrity shall be verified (valid JPEG headers/JSON parse).	Muhammed

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Test ID	Requirement(s)	Method / Procedure (incl. Acceptance)	Resp.
TN-NFR-SW-02	*HTTP/REST API over LTE shall upload events (JPEG/WAV/JSON) to cloud endpoints.*	An incident upload shall be executed from gateway to cloud using the REST endpoint. Server response codes and upload acknowledgments shall be logged. **Acceptance:** HTTP response codes shall indicate success and uploaded objects shall be retrievable from the cloud/app.	İsmail
TN-NFR-SW-03	*Ethernet link shall provide network connectivity over ethernet socket.*	Connectivity shall be maintained under continuous upload traffic for 3 minutes. **Acceptance:** session shall remain active and cloud reachability shall be maintained without repeated reconnects.	Yusuf

3.7 User Interface Non-Functional Tests

Table 2: Non-Functional Tests for Mobile Application User Interface Modules

Test ID	Requirement(s)	Method / Procedure (incl. Acceptance)	Resp.
TN-NFR-UI-01	*Login shall be performed securely; redirection to main page shall occur after authentication.*	Valid credentials shall be entered and login shall be executed. Invalid credentials shall be entered and rejection behavior shall be observed. **Acceptance:** Successful login shall redirect to the main page; invalid login shall not grant access and an error indication shall be shown.	İrem

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Test ID	Requirement(s)	Method / Procedure (incl. Acceptance)	Resp.
TN-NFR-UI-02	*Incident/notification feed shall be accessible and information shall be visible in relevant screens.*	The notification page shall be opened and recent alerts shall be displayed. **Acceptance:** Alerts shall be listed consistently and filters shall update the displayed list without crash or UI freeze.	İrem
TN-NFR-UI-03	*Recordings and media playback shall be accessible from the recordings module.*	A stored incident media item shall be opened from the recordings page. Playback shall be started and stopped. **Acceptance:** Media shall be retrievable and playback controls shall function without application crash.	İrem
TN-NFR-UI-04	*Map view shall display node locations and allow zoom and node-specific inspection.*	The map page shall be opened and node markers shall be displayed. **Acceptance:** Node locations shall be visible and interactions (zoom/select) shall update the view correctly.	İrem
TN-NFR-UI-05	*Account management pages shall be reachable and logout shall terminate session safely.*	The account page and edit info page shall be opened. Logout shall be performed. **Acceptance:** Session shall be terminated and the login screen shall be shown; returning to protected pages without login shall be prevented.	İrem

References

[1] Y. S. Gürkök, M. Gölcük, İ. Özgenç, and Z. İ. İpek, *FarmGate Sentinel Product Specification*, EEE495 Design II, Bilkent University, Ankara, Türkiye, 2025.